

RFID ECRT APPLICATION NOTE

App Lifecycle & RFID Reader Management

Zebra RFID SDK on TC701 with RFD40 / RFD8500 Readers

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How the sample app drives the Zebra RFID SDK across the Android activity lifecycle: SDK init, reader discovery, suspend/background disconnect, resume reconnect, and USB cable plug-in / unplug handling.

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This document describes how the sample app drives the Zebra RFID SDK across the Android activity lifecycle: **SDK initialization**, **reader discovery**, **background disconnect / dispose**, **suspend disconnect**, **resume reconnect**, and **USB cable plug-in / unplug** (reader disappear / reappear + auto-reconnect).

The two collaborating classes are:

- **MainActivity** — owns the UI and the Android lifecycle callbacks. It forwards lifecycle transitions to the handler.
- **RFIDHandler** — owns the **Readers / RFIDReader** SDK objects and all connect / disconnect / dispose logic.

Design principle

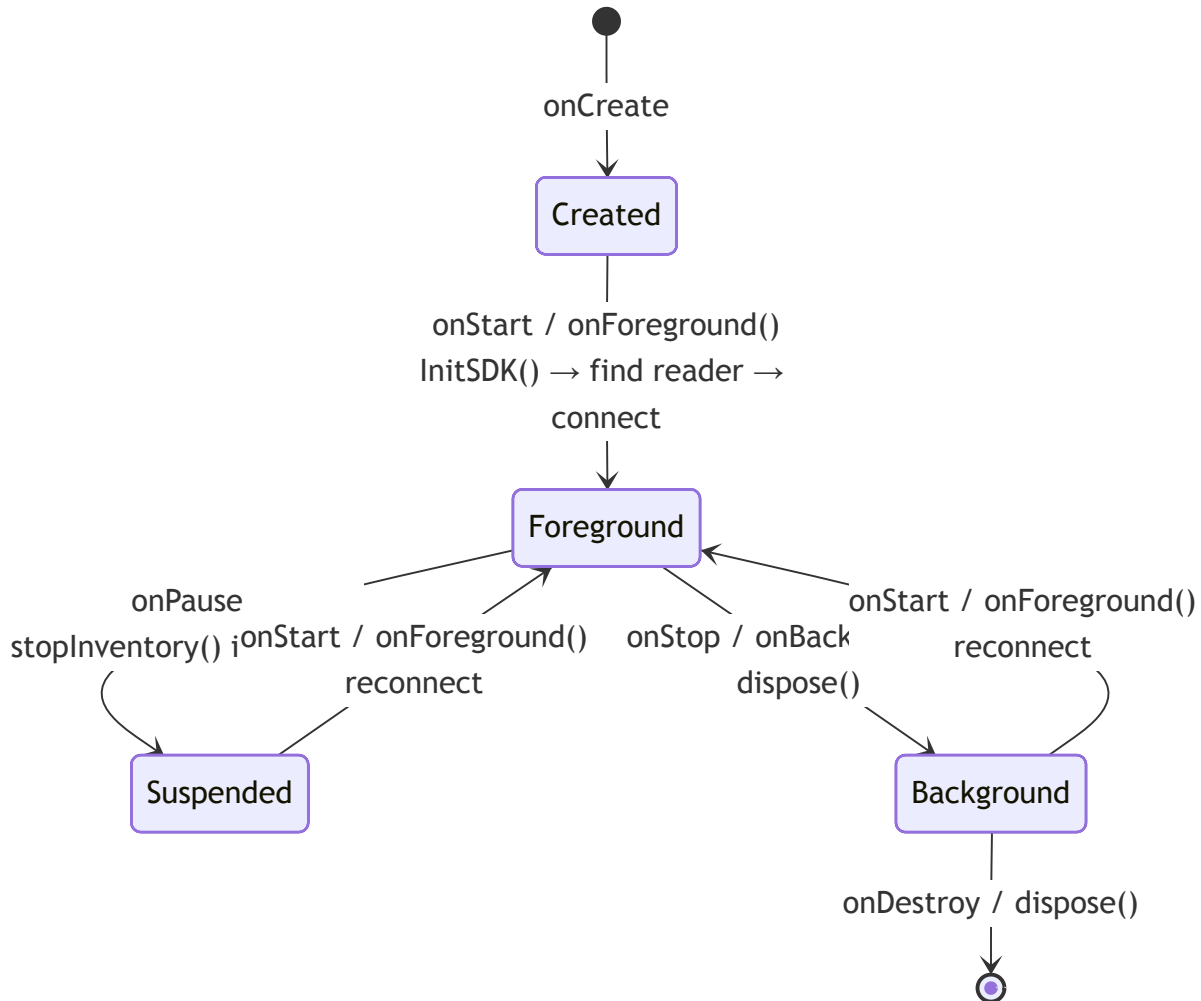
The app treats **foreground = live SDK** and **background = no SDK**:

- When the app becomes visible it fully **initializes the SDK, finds the reader, and connects**.
- When the app leaves the screen it **disconnects, cleans up, and disposes** the SDK so no reader session or native resources are held while backgrounded.

This keeps the reader available to other apps while this one is not in use, and guarantees a clean, well-defined reconnect every time the user returns.

Lifecycle map

Android callback (MainActivity)	Handler call (RFIDHandler)	Effect
onCreate	<i>(none — only wires views & requests permission)</i>	Prepares UI, sets rfidPermissionReady
onStart (foreground)	onForeground(this) → onCreate(...) → InitSDK()	Init SDK, find reader, connect
onPostResume	<i>(none — just refreshes title/status)</i>	Clears status buffer, updates title
onPause (partial suspend)	stopInventory() (only if reading)	Stops an in-flight inventory
onStop (background)	onBackground() → dispose()	Disconnect + clean + dispose
onDestroy	onDestroy() → dispose()	Final cleanup



Events that are **independent of the activity lifecycle** — reader-initiated disconnects and USB cable plug-in / unplug — are covered in sections 6 and 7.

1. Permission gate (before any SDK use)

The reader transports need Bluetooth permissions on Android 12+. `onCreate` does **not** initialize the SDK directly; it only records whether permission is available via `rfidPermissionReady`. Actual initialization is always driven by `onStart` (see section 2).

```
// MainActivity.onCreate(...)
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
    if (ContextCompat.checkSelfPermission(this, Manifest.permission.BLUETOOTH_CONNECT) != PackageManager.PERMISSION_GRANTED) {
        requestPermissionLauncher.launch(new String[]{Manifest.permission.BLUETOOTH_SCAN, Manifest.permission.BLUETOOTH_CONNECT})
    } else {

```

```

        rfidPermissionReady = true;
    }
} else {
    rfidPermissionReady = true;
}

```

Because the permission dialog resolves **after** `onStart` has already run, the grant callback must kick off initialization itself:

```

// MainActivity - permission result callback
private final ActivityResultLauncher<String[]> requestPermissionLauncher =
    registerForActivityResult(new ActivityResultContracts.RequestMultiplePermissions(),
        Boolean bluetoothConnect = result.getDefault(Manifest.permission.BLUETOOTH_CONNECT)
        if (Boolean.TRUE.equals(bluetoothConnect)) {
            rfidPermissionReady = true;
            rfidHandler.onForeground(this); // permission arrived after onStart → initialize
        } else {
            Toast.makeText(this, R.string.permission_not_granted, Toast.LENGTH_SHORT).show();
        }
    });

```

2. Foreground: init SDK, find reader, connect

`onStart` fires every time the activity becomes visible (cold start and every return from background). Guarded by the permission flag, it asks the handler to enter the foreground state.

```

// MainActivity.onStart()
@Override
protected void onStart() {
    super.onStart();
    Log.d(TAG, "onStart");
    if (rfidPermissionReady) {
        rfidHandler.onForeground(this);
    }
}

```

`onForeground` re-establishes handler references and (re)creates the `ToneGenerator` if it was released by a prior `dispose()`, then calls `InitSDK()`:

```

// RFIDHandler
void onForeground(MainActivity activity) {
    onCreate(activity);
}

```

```

void onCreate(MainActivity activity) {
    context = activity;
    textView = activity.statusTextViewRFID;
    scannerList = new ArrayList<>();
    if (toneGenerator == null) { // recreated after a background dispose
        try {
            toneGenerator = new ToneGenerator(AudioManager.STREAM_NOTIFICATION, 100);
        } catch (Exception e) {
            Log.e(TAG, "Failed to initialize ToneGenerator", e);
        }
    }
    InitSDK();
    context.updateTitleWithSdkVersion(getVersionInfo());
}

```

2a. InitSDK() — create the Readers instance or reconnect

If the SDK was disposed (`readers == null`), it shows progress, sets a transient status (`Connecting...` on first launch, `Reconnecting...` afterwards) and builds a fresh `Readers` instance on a background thread. If the instance still exists, it simply reconnects.

```

// RFIDHandler.InitSDK()
private void InitSDK() {
    Log.d(TAG, "InitSDK");
    IRFIDLogger.getLogger("sample app").EnableDebugLogs(true);
    if (readers == null) {
        context.showProgress(true);
        context.setRfidStatus(hasConnectedBefore ? "Reconnecting..." : "Connecting...");
        executorService.execute(this::createInstance);
    } else {
        performConnect();
    }
}

```

2b. createInstance() — find the reader across transports

Reader discovery tries a series of transports (`BLUETOOTH` → `SERVICE_SERIAL` → `SERVICE_USB` → `QC_SERIAL` → `RE_SERIAL`), stopping as soon as `GetAvailableRFIDReaderList()` returns a non-empty list.

```

// RFIDHandler.createInstance() – transport probing (abridged)
readers = new Readers(context, ENUM_TRANSPORT.BLUETOOTH);
availableRFIDReaderList = readers.GetAvailableRFIDReaderList();

if (availableRFIDReaderList.isEmpty()) {
    readers.setTransport(ENUM_TRANSPORT.SERVICE_SERIAL);
}

```

```

        transport = ENUM_TRANSPORT.SERVICE_SERIAL;
        availableRFIDReaderList = readers.GetAvailableRFIDReaderList();
    }
    // ... SERVICE_USB, QC_SERIAL, RE_SERIAL fallbacks follow the same pattern ...

    if (!availableRFIDReaderList.isEmpty()) {
        Log.d(TAG, "Found Reader, Size=" + availableRFIDReaderList.size());
        Log.d(TAG, "Init OK for Transport = " + transport.toString());
    }

```

The result is then applied on the UI thread. If discovery failed or no reader was found, the UI is reset to the disconnected state; otherwise it proceeds to connect.

```

// RFIDHandler.createInstance() - result handling
context.runOnUiThread() -> {
    if (finalException != null) {
        context.sendToast(context.getString(R.string.status_failed_get_readers) + "\n" + fi
        readers = null;
        context.showProgress(false);
        context.setConnectionButtons(false);
    } else if (availableRFIDReaderList.isEmpty()) {
        context.sendToast(context.getString(R.string.status_no_readers));
        textView.setText("No Reader Found\r\nPower on Reader or Attach");
        readers = null;
        context.showProgress(false);
        context.setConnectionButtons(false);
    } else {
        connectReader();
    }
});

```

2c. GetAvailableReader() — pick the reader device

When connecting, the handler selects the reader: if exactly one is available it uses it, otherwise it matches by `readerName`.

```

// RFIDHandler.GetAvailableReader() (abridged)
readers.attach(this);
ArrayList<ReaderDevice> availableReaders = readers.GetAvailableRFIDReaderList();
if (availableReaders != null && !availableReaders.isEmpty()) {
    availableRFIDReaderList = availableReaders;
    if (availableRFIDReaderList.size() == 1) {
        readerDevice = availableRFIDReaderList.get(0);
        reader = readerDevice.getRFIDReader();
    } else {
        for (ReaderDevice device : availableRFIDReaderList) {
            if (device.getName().startsWith(readerName)) {

```

```

        readerDevice = device;
        reader = readerDevice.getRFIDReader();
    }
}
if (impinjExtensions == null)
    impinjExtensions = new ImpinjExtensions(reader);
}

```

2d. `connectReader()` / `handleConnect()` — establish the session

```

// RFIDHandler.connectReader()
private synchronized void connectReader() {
    if (!isReaderConnected()) {
        context.showProgress(true);
        executorService.execute(this::connectionTask);
    }
}

```

`connectionTask()` runs discovery + connect, handles the `RFID_READER_REGION_NOT_CONFIGURED` case, configures the reader, then updates the UI (status text + CONNECT/DISCONNECT button state) with the final result.

```

// RFIDHandler.connectionTask() – final UI update
final String result = res;
context.runOnUiThread(() -> {
    textView.setText(result);
    context.UpdateUI_statusTextViewRFID(result);
    context.showProgress(false);
    context.setConnectionButtons(result != null && result.startsWith("Connected"));
});

```

```

// RFIDHandler.handleConnect() – actual connect call
if (!reader.isConnected()) {
    reader.connect();
}
if (reader.isConnected()) {
    hasConnectedBefore = true; // flips future status text to "Reconnecting"
    playTone(ToneGenerator.TONE_SUP_CONFIRM); // confirmation beep
    return "Connected: " + reader.getHostName();
}

```

3. Suspend: onPause stops an in-flight inventory

`onPause` is a *partial* suspend (e.g. a dialog or the recents overlay appears). The app does **not** disconnect here; it only stops an active inventory so no read is left running while the activity is not resumed.

```
// MainActivity.onPause()
@Override
protected void onPause() {
    if (rfidHandler.isIsReading())
        rfidHandler.stopInventory();
    super.onPause();
    Log.d(TAG, "onPause");
}
```

4. Background: onStop disconnects, cleans up, disposes

When the activity is no longer visible, the app tears the SDK down completely.

```
// MainActivity.onStop()
@Override
protected void onStop() {
    rfidHandler.onBackground();
    super.onStop();
    Log.d(TAG, "onStop");
}
```

```
// RFIDHandler
void onBackground() {
    dispose();
}
```

`dispose()` disconnects the reader, releases the `ToneGenerator`, clears the Impinj extension, nulls the reader, and disposes the `Readers` instance so `readers == null` — which is exactly the condition `InitSDK()` uses on the next foreground to know it must rebuild the SDK from scratch.

```
// RFIDHandler.dispose()
private synchronized void dispose() {
    disconnectInternal();
    try {
        if (toneGenerator != null) {
            toneGenerator.release();
            toneGenerator = null;
        }
    }
}
```

```

    }
    impinjExtensions = null;
    if (reader != null) {
        reader = null;
        readers.Dispose();
        readers = null;
    }
} catch (Throwable e) {
    e.printStackTrace();
}
}
}

```

`disconnectInternal()` performs the low-level teardown: it removes the event listener, terminates the scanner session, and disconnects the reader, swallowing the SDK exceptions that can occur while tearing down.

```

// RFIDHandler.disconnectInternal() (abridged)
if (reader != null) {
    if (eventHandler != null)
        reader.Events.removeEventListener(eventHandler);
    if (sdkHandler != null) {
        sdkHandler.dcssdkTerminateCommunicationSession(scannerID);
        scannerList = null;
    }
    try {
        reader.disconnect();
    } catch (OperationFailureException ofe) {
        Log.d(TAG, "OperationFailureException ofe=" + ofe.getVendorMessage());
    } catch (Throwable e1) {
        Log.d(TAG, "Exception/Error e=" + e1.getMessage());
    }
    context.sendToast("Disconnecting reader");
}
}

```

5. Resume: reconnect

Returning from background (or from a partial suspend) fires `onStart` again, which calls `onForeground(this)` → `InitSDK()`. Because `dispose()` set `readers == null`, `InitSDK()` rebuilds the `Readers` instance, re-runs transport discovery, and reconnects — showing “**Reconnecting...**” (since `hasConnectedBefore` is now `true`) followed by “**Connected:** ”.

This is the same code path as the initial foreground connect (section 2); the only difference is the status wording driven by `hasConnectedBefore`.

6. Unsolicited disconnects (reader-initiated)

Independent of the activity lifecycle, the reader can drop the link (power off, out of range, cable removed). The SDK reports this via a status event, which routes to the same `handleDisconnect()` cleanup used elsewhere.

```
// RFIDHandler.EventHandler.eventStatusNotify(...)
if (rfidStatusEvents.StatusEventData.getStatusEventType() == STATUS_EVENT_TYPE.DISCONNECTION) {
    executorService.execute(RFIDHandler.this::handleDisconnect);
}
```

The reader-appear/disappear callbacks (from `Readers.RFIDReaderEventHandler`) provide auto-reconnect when the reader shows up on a transport again:

```
// RFIDHandler
@Override
public void RFIDReaderAppeared(ReaderDevice readerDevice) {
    Log.d(TAG, "RFIDReaderAppeared " + readerDevice.getName());
    context.sendToast("Reader appeared: " + readerDevice.getName());
    if (reader != null && reader.isConnected()) {
        context.sendToast("Reader in USB host mode");
        if (readerDevice.getName().equals(reader.getHostName()))
            performDisconnect(); // same reader re-announced → drop the stale session
    }
    connectReader(); // (re)connect to the reader that just appeared
}

@Override
public void RFIDReaderDisappeared(ReaderDevice readerDevice) {
    Log.d(TAG, "RFIDReaderDisappeared " + readerDevice.getName());
    context.sendToast("Reader disappeared: " + readerDevice.getName());
    // Only tear down if the reader that vanished is the one we are connected to.
    if (reader != null && readerDevice.getName().equals(reader.getHostName()))
        performDisconnect();
}
```

7. USB cable plug-in / unplug (TC701 + RFD40 in a cradle)

On the TC701 the RFD40 reader is reached over a **serial transport** (`SERVICE_SERIAL`), *not* over the device's USB port. Plugging a USB cable into the TC701 (for example, to charge or to connect to a host PC) flips the USB role and **preempts the serial link to the reader**, so the SDK raises a `RFIDReaderDisappeared` event. Removing the cable frees the link again and the SDK raises a `RFIDReaderAppeared` event.

The handler is written so this happens automatically, with no user action:

Physical event	SDK callback	Handler behavior
USB cable plugged in → reader link preempted	RFIDReaderDisappeared (and/or DISCONNECTION_EVENT)	RFIDReaderDisappeared calls <code>performDisconnect()</code> for the matching host, which runs <code>handleDisconnect()</code> → <code>disconnectInternal()</code> , plays the “phone drop” tone, greys out the buttons and shows “ Disconnected ”. If a <code>DISCONNECTION_EVENT</code> also fires it funnels into the same synchronized <code>handleDisconnect()</code> .
USB cable unplugged → reader link restored	RFIDReaderAppeared	<code>connectReader()</code> re-runs discovery + connect and shows “ Connected: <host> ” again.

Why the disconnect is guarded

`RFIDReaderDisappeared` only tears down when the reader that vanished is the one this app is actually connected to:

```
if (reader != null && readerDevice.getName().equals(reader.getHostName()))
    performDisconnect();
```

Two reasons for the guard:

1. **Null-safety.** After a previous teardown `reader` can already be `null`; dereferencing `reader.getHostName()` unconditionally would throw a `NullPointerException`. The `reader != null` check prevents that.
2. **Ignore unrelated readers.** A disappear event for a *different* device must not drop our live session; the host-name match scopes the teardown to our own reader.

`performDisconnect()` hands off to the `synchronized handleDisconnect()`, so if a `DISCONNECTION_EVENT` (section 6) races with the disappear callback the two are serialized rather than interleaved.

The “same reader reappears while still connected” guard

When the cable is removed, the reader can be re-announced *before* the stale session from the previous connection has been fully cleaned up. The guard inside `RFIDReaderAppeared` handles that case:

```
if (reader != null && reader.isConnected()) {
    context.sendToast("Reader in USB host mode");
}
```

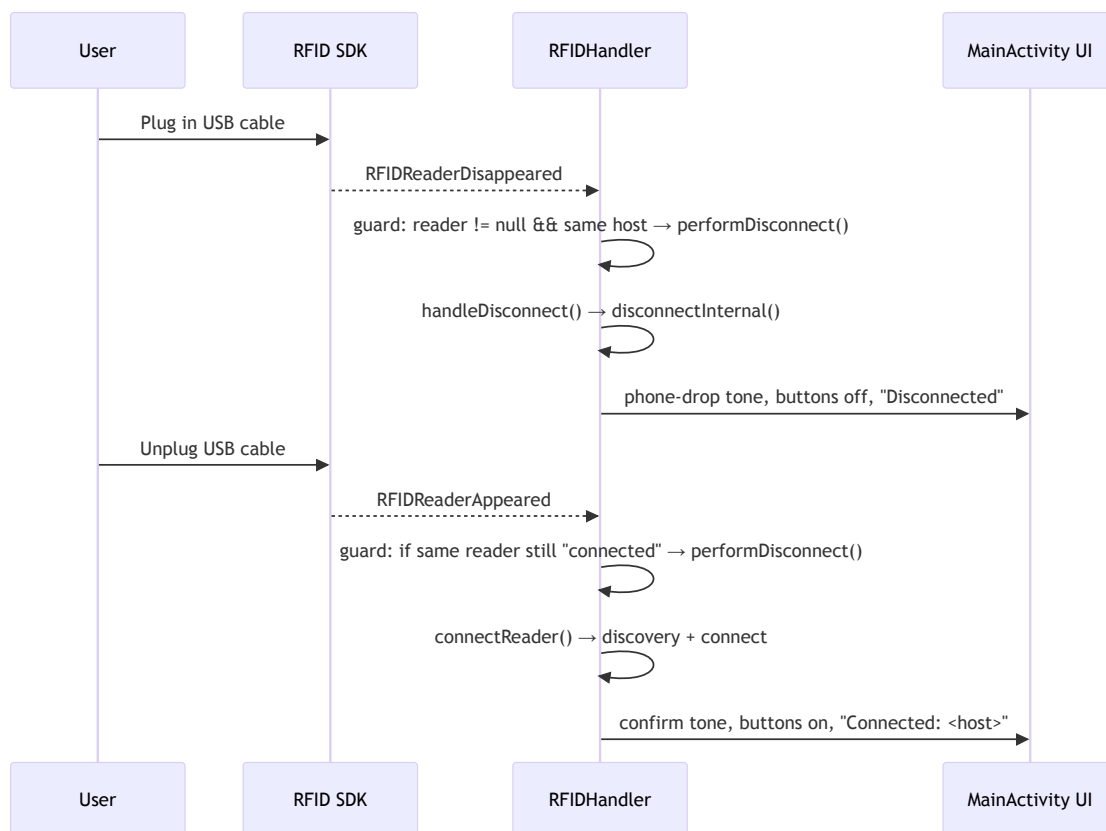
```

if (readerDevice.getName().equals(reader.getHostName()))
    performDisconnect(); // drop the stale session before reconnecting
}
connectReader();

```

If the *same* host name reappears while we still think we are connected, the handler disconnects first (clearing the dead session) and then `connectReader()` establishes a fresh one. If a *different* reader appears, it simply connects to it.

Sequence



Note (edge cases to watch): - If `connectReader()` fires before the serial port is fully released after unplugging, the connect can fail. `connectReader()` is idempotent (if `!isReaderConnected()`), so a subsequent `RFIDReaderAppeared` / `RFIDReaderAppeared`-triggered retry will reconnect cleanly. - Debounce rapid plug/unplug cycles if you see overlapping connect/disconnect toasts — each physical transition can emit paired appear/disappear events.

`handleDisconnect()` is the single funnel for every disconnect (lifecycle,

user-initiated, or reader-initiated). It tears down the session, plays the "phone drop" beep, and updates the UI to the disconnected state.

```
```java
// RFIDHandler.handleDisconnect()
private synchronized void handleDisconnect() {
 disconnectInternal();
 playPhoneDropTone();
 context.setConnectionButtons(false);
 context.UpdateUI_statusTextViewRFID("Disconnected");
}
```

---

## Summary

---

- **Init SDK** — `onStart` → `onForeground()` → `InitSDK()` builds the `Readers` instance on a background thread (gated by Bluetooth permission).
- **Find reader** — `createInstance()` probes transports (BLUETOOTH → SERVICE\_SERIAL → SERVICE\_USB → QC\_SERIAL → RE\_SERIAL); `GetAvailableReader()` selects the device.
- **Suspend disconnect** — `onPause` only stops an active inventory; the connection is kept.
- **Background disconnect** — `onStop` → `onBackground()` → `dispose()` fully disconnects, cleans up, and releases the SDK (`readers == null`).
- **Resume reconnect** — `onStart` runs again; since `readers == null`, `InitSDK()` rebuilds and reconnects, showing “Reconnecting...” → “Connected”.
- **Reader-initiated disconnects** and **auto-reconnect** funnel through `handleDisconnect()` / `connectReader()`, sharing the lifecycle code paths.
- **USB cable plug-in / unplug** — `RFIDReaderDisappeared` actively disconnects the matching reader (guarded by `reader != null` + host-name match); `RFIDReaderAppeared` reconnects it. See section 7.